Supplementary Material for "Complementary-Filtered Residual World Models for Receding-Horizon Control of Nonlinear Systems"

Anonymous Author(s)

APPENDIX A
BENCHMARK SPECIFICATION

Table I records the simulator time step and nominal parameterization; Table II gives every sampled training and OOD interval. All OOD intervals are non-overlapping on every named shifted parameter. These ranges define the fixed benchmark.

The action generator is a bounded first-order filtered process. At step t, a Gaussian innovation with scale 0.55 times the system action limit is combined with the previous action using coefficients 0.18 and 0.82, then clipped. Limits are 1.5, 1.8, 1.2, 2.2, and 8.0 for the systems in Table I. This produces persistent excitation independently of the simulated state and defines the control-input distribution.

Train/validation/test separation occurs before window construction. A root NumPy `SeedSequence` spawns distinct split generators, which in turn spawn one generator per trajectory. The OOD test has a separate root seed. Observation corruption uses another recorded seed after state generation. Consequently, changing an observation-noise level does not change the underlying clean ID trajectory, and all methods see exactly the same corruption draw within a condition.

APPENDIX B
CONFIRMATORY GRID

The prespecified grid comprises 11 methods, eight seeds (41, 53, 67, 79, 83, 97, 109, and 113), five systems, and three test regimes. It therefore contains 440 fitted models, 1,320 aggregate evaluation cells, and 126,720 trajectory-level rows. Each seed–system dataset contains 256 training, 64 validation, and 96 test trajectories of 72 time steps. Models receive eight context steps and predict 25 steps open loop. Training observations use noise equal to 3% of the fixed training-state scale and a nominal 10% independent missing probability; the corrupt regimes use 8% observation noise and a nominal 25% probability. The first time point of each generated trajectory is forced observed in every channel, with independent masks sampled at all subsequent trajectory times; a later sliding-window context can begin with an imputed value. ID-clean and ID-corrupt evaluations share trajectory identifiers; the OOD trajectory set is disjoint.

The five broad comparators are compact, protocol-matched RSSM-style, SimNorm-style, Transformer-style, PGNN-style, and PG-DMM-style transitions. Five prespecified CFR sensitivities test gains 0, 0.25, 0.75, and 1, plus a one-step-trained gain-0.5 model. CFR-WM retains the unused learned-gain-head parameters to maintain identical parameter counts across fixed-gain comparisons.

All methods ingest normalized context values, masks, log-one-plus elapsed times, and actions. Except for the Transformer-style encoder, context is encoded by a GRU; open-loop recurrent updates use a GRUCell. Every MLP has two hidden linear layers with SiLU activation. Table III lists the remaining architecture-specific choices.

For CFR-WM, observed context channels use gain 0.5 and missing channels use gain 0 exactly. The learned gain head remains instantiated but is inactive. The four gain sensitivities change only the observed-channel gain; the one-step sensitivity changes only the likelihood horizon.

APPENDIX C
HYPERPARAMETER SEARCH

The same six learning-rate/weight-decay trials were run on development seed 11 for every system–baseline pair (Table IV). The stored reliability-loss weight is shown because it is part of the generic training configuration, but it is inactive for all confirmatory methods in this paper. Each baseline uses the validation-NLL winner in Table V; CFR-WM and all CFR sensitivities inherit the PGNN-style winner for the same system.

APPENDIX D
CONFIRMATORY ESTIMANDS AND DECISIONS

H1 is the sole confirmation gate. It asks whether CFR-WM improves fixed-training-scale NRMSE-25 over the preselected PGNN-style comparator by at least 5%. Within each seed, PGNN-style and CFR-WM are averaged with equal weight over all 15 fixed system–regime cells; the primary point estimate is the arithmetic mean of the eight seed-specific relative improvements. H1 requires finite metrics, a 95% bootstrap lower endpoint of at least 5%, and a one-sided exact sign-flip $p < 0.05$ for the 5% margin.

H2 is a hierarchically gated robustness claim. Conditional on H1, the combined corrupt and parameter-shifted regimes must satisfy the same 5% improvement rule, while ID-clean performance must be noninferior under a signed relative

TABLE I
CONTROLLED-SYSTEM DEFINITIONS. u IS SCALAR IN EVERY SYSTEM.

System	State	Δt	Nominal parameters
Damped oscillator	$(q, \dot{q})$	0.050	$m = 1.0,\ k = 1.6,\ c = 0.18$
Forced Duffing	$(q, \dot{q})$	0.040	$\alpha = -1.0,\ \beta = 1.0,\ c = 0.25$
Controlled Van der Pol	$(q, \dot{q})$	0.035	$\mu = 1.2,\ g_u = 1.0$
Torque pendulum	$(\sin\theta, \cos\theta, \dot{\theta})$	0.040	$g = 9.81,\ l = 1.0,\ m = 1.0,\ c = 0.12$
Cart–pole	$(x, \dot{x}, \theta, \dot{\theta})$	0.025	$g = 9.81,\ m_c = 1.0,\ m_p = 0.1,\ l/2 = 0.5,\ c = 0.04$

TABLE II
COMPLETE INDEPENDENTLY SAMPLED PHYSICAL-PARAMETER INTERVALS. SYMBOLS FOLLOW TABLE I; g_u IS THE VAN DER POL CONTROL GAIN.

System	Training interval	OOD interval
Damped oscillator	$m \in [0.90, 1.10],\ k \in [1.40, 1.80],\ c \in [0.14, 0.22]$	$m \in [1.25, 1.45],\ k \in [2.05, 2.35],\ c \in [0.26, 0.34]$
Forced Duffing	$\alpha \in [-1.10, -0.90],\ \beta \in [0.90, 1.10],\ c \in [0.20, 0.30]$	$\alpha \in [-0.75, -0.60],\ \beta \in [1.25, 1.45],\ c \in [0.34, 0.44]$
Controlled Van der Pol	$\mu \in [1.00, 1.40],\ g_u \in [0.90, 1.10]$	$\mu \in [1.65, 1.95],\ g_u \in [1.25, 1.45]$
Torque pendulum	$g \in [9.65, 9.95],\ l \in [0.90, 1.10],\ m \in [0.90, 1.10],\ c \in [0.08, 0.16]$	$g \in [10.25, 10.55],\ l \in [1.25, 1.45],\ m \in [1.25, 1.45],\ c \in [0.20, 0.28]$
Cart–pole	$g \in [9.65, 9.95],\ m_c \in [0.90, 1.10],\ m_p \in [0.085, 0.115],\ l/2 \in [0.45, 0.55],\ c \in [0.02, 0.06]$	$g \in [10.25, 10.55],\ m_c \in [1.25, 1.45],\ m_p \in [0.135, 0.160],\ l/2 \in [0.65, 0.78],\ c \in [0.075, 0.105]$

TABLE III
EXACT COMPACT TRANSITION ARCHITECTURES USED IN THE CONFIRMATORY GRID. THE BOUNDED CORRECTION IS $0.25\tanh(\cdot)$ AND ALL FORECAST COVARIANCES ARE DIAGONAL.

Method	Width	Context representation	Forecast transition and variance
RSSM-style	80	GRU; sigmoid feature scaling	data-only MLP mean/log-variance; GRUCell rollout
SimNorm-style	80	GRU; eight-group simplex projection	data-only MLP mean/log-variance; GRUCell rollout
Transformer-style	48	two causal layers, four heads, FF width 96, GELU, dropout 0	data-only MLP mean/log-variance; GRUCell rollout
PGNN-style	84	GRU	nominal one-substep RK4 plus deterministic residual MLP; learned constant log-variance
PG-DMM-style	84	GRU	nominal one-substep RK4 plus residual mean/log-variance MLP
CFR-WM and gain sensitivities	72	GRU states plus mask-gated fixed-gain context update	nominal one-substep RK4 plus residual mean/log-variance MLP

TABLE IV
SIX-TRIAL OPTIMIZATION GRID.

Trial	Learning rate	Weight decay	Stored λ_{rel}
0	3×10^{-4}	10^{-6}	0.1
1	6×10^{-4}	10^{-5}	0.2
2	10^{-3}	10^{-5}	0.2
3	1.5×10^{-3}	10^{-4}	0.3
4	2×10^{-3}	10^{-5}	0.4
5	3×10^{-3}	10^{-4}	0.2

margin of -5%. Both components require their corresponding bootstrap lower endpoint to meet or exceed the margin and a one-sided exact sign-flip $p < 0.05$. If H1 fails, H2 is not supported irrespective of its component estimates.

The bootstrap uses 5,000 replicates. Each replicate resamples the eight seed clusters globally with replacement and then resamples paired trajectory identifiers within every retained seed–system block. A single sampled identifier vector is reused for ID-clean and ID-corrupt repeated measurements, whereas the disjoint OOD identifiers are sampled separately. All five systems and three regimes remain in every replicate, and baseline, candidate, and their relative ratio are recomputed within the replicate. Exact tests enumerate all 2^8 seed-level sign flips and therefore rely on a seed-effect symmetry assumption. The five broad-comparator contrasts and the five sensitivity contrasts form separate Holm-adjusted families. The 15 system–regime contrasts are descriptive and carry no cell-level significance claim.

APPENDIX E
SECONDARY ANALYSIS PROTOCOL

The classical baseline applies the same normalized nominal step and state projection as CFR-WM. Its unfiltered version starts from the final causally imputed context state; its filtered version applies gain 0.5 to each observed channel throughout the eight-step context. Both then use nominal dynamics only for all 25 forecast steps. The residual-scale analysis loads every trained CFR-WM checkpoint and changes only the multiplier on $\tanh(\mathbf{r})$ to 0, 0.125, 0.25, or 0.5. The stored 0.25 aggregate is reproduced with maximum absolute difference 0.00e+00.

The structured-outage audit starts from clean ID windows and makes all channels missing for context pair 3–4, 5–6, or 7–8. The previous observation is carried forward and elapsed time is set to one and two steps. Results average the three positions, five systems, and fixed cells within each of the eight seeds. The projection audit selects missing sine/cosine entries in the corrupt ID and corrupt OOD pendulum contexts and measures the absolute difference between the preprojection and projected normalized states.

Stored latency is computed separately for every method, seed, system, and regime. One batch contains all 96 evaluation trajectories. Timing begins before the batch tensors move from CPU to GPU, synchronizes CUDA after the 25-step rollout, and excludes data-loader iteration and metric computation. The value is divided by 96 before seed-balanced aggregation. It is a batched throughput measure.

External uncertainty is recomputed from the three held-

TABLE V
SELECTED TRIAL INDEX BY SYSTEM AND PROTOCOL-MATCHED BASELINE.

System	RSSM-style	SimNorm-style	Transformer-style	PGNN-style	PG-DMM-style
Damped oscillator	3	2	4	5	1
Forced Duffing	5	4	5	5	5
Controlled Van der Pol	5	5	5	5	0
Torque pendulum	5	5	5	5	4
Cart–pole	5	5	5	5	5

out flight predictions for all five final seeds of CFR-WM, the unfiltered residual model, and the data- only recurrent model. Flight-level NLL, nominal 90% marginal coverage, mean interval width, and central-interval calibration error are first given equal flight weight and then summarized across seeds. Recomputed prediction means match the stored means with maximum absolute difference 0.00e+00.

APPENDIX F
REPRODUCIBILITY CHECKS

The reproducibility contract records the method list, seeds, data design, training configuration, decision rules, and SHA-256 hashes of the training, evaluation, analysis, and tuning artifacts. Analysis requires the complete grid, finite metrics, trajectory pairing, disjoint OOD identifiers, 40 distinct seed–system data hashes, and matching stored provenance. The top-level independent units are the eight stochastic training/data realizations, with trajectories nested within them. Statistical inference concerns the prespecified finite benchmark and variation across those seed realizations.

APPENDIX G
COMPLETE DESCRIPTIVE CELL CONTRASTS

Table VI reports every prespecified CFR-WM–PGNN-style system–regime contrast. Absolute intervals come from the paired nested bootstrap. Relative changes use PGNN-style as the denominator, with negative values favoring CFR-WM. These fixed-cell estimates are descriptive.

APPENDIX H
CLOSED-LOOP PROTOCOL AND COMPLETE AGGREGATES

The closed-loop study loads the frozen CFR-WM and PGNN-style checkpoint for each seed–system pair. The nominal-last and filter-plus-nominal interfaces use the CFR-WM checkpoint's normalization buffers and nominal step but do not use its residual. Equality of the CFR-WM and PGNN normalization buffers is checked before each comparison. Table VII gives the complete controller and condition aggregates.

The plant and controller integration uses the benchmark sampling and four-substep Runge–Kutta functions. The episode receipt stores controller, seed, system, condition, episode index, realized metrics, initial state, plant parameters, and planner seed. Each controller quartet must have identical initial state, plant parameters, and planner seed. ID-clean and ID-corrupt episodes must also have identical plant draws. Asset generation rejects a missing or duplicate identity, a nonfinite metric, a changed checkpoint or source hash, or disagreement with the recomputed aggregates and inference.

Figure 1 reports the same aggregate loss as the main paper and all 15 descriptive system–condition CFR-WM contrasts against PGNN-style. Positive percentages favor CFR-WM.

APPENDIX I
MEASURED-FLIGHT PROTOCOL AND FULL RESULTS

The external data set contains timestamped position, linear velocity, Euler attitude, angular rates, and four motor commands. The reduced nominal prior uses mass-normalized thrust, linear drag, an Euler-angle kinematic approximation, inertia-scaled motor moments, and angular damping. The learned residual retains the same 0.25 normalized bound used in the controlled benchmark. Training uses only the designated training flight family; development selects the learning rate from $\{10^{-3}, 3 \times 10^{-3}\}$ on a separate seed. Five final seeds are then fit once. Sliding evaluation windows do not cross flight boundaries.

Table VIII reports full-state NRMSE for every method, context condition, and horizon. Table IX gives position and velocity errors. The nominal prior has no training-seed variation, so its reported SD is zero.

The probabilistic models use diagonal Gaussian marginals. The full external uncertainty diagnostics are shown in Table X.

TABLE VI
ALL 15 DESCRIPTIVE CFR-WM VERSUS PGNN-STYLE NRMSE-25 CONTRASTS.

System	Regime	CFR–PGNN-style [95% CI]	Relative change, %
Cartpole	ID clean	+0.0163 [+0.0103, +0.0225]	+47.33
Cartpole	ID corrupt	-0.0207 [-0.0281, -0.0136]	-13.33
Cartpole	OOD + corrupt	-0.0111 [-0.0194, -0.0027]	-5.66
Controlled Van Der Pol	ID clean	+0.0059 [-0.0022, +0.0149]	+19.36
Controlled Van Der Pol	ID corrupt	-0.0318 [-0.0381, -0.0250]	-32.39
Controlled Van Der Pol	OOD + corrupt	-0.0083 [-0.0148, -0.0016]	-4.74
Damped Oscillator	ID clean	-0.0011 [-0.0041, +0.0019]	-2.41
Damped Oscillator	ID corrupt	-0.0291 [-0.0341, -0.0244]	-32.12
Damped Oscillator	OOD + corrupt	-0.0257 [-0.0295, -0.0223]	-27.86
Forced Duffing	ID clean	+0.0094 [+0.0028, +0.0168]	+24.37
Forced Duffing	ID corrupt	-0.0295 [-0.0342, -0.0249]	-30.24
Forced Duffing	OOD + corrupt	+0.0094 [+0.0011, +0.0174]	+3.90
Torque Pendulum	ID clean	-0.0046 [-0.0082, -0.0005]	-8.59
Torque Pendulum	ID corrupt	-0.0378 [-0.0436, -0.0316]	-34.80
Torque Pendulum	OOD + corrupt	+0.0110 [+0.0055, +0.0165]	+5.94

TABLE VII
CLOSED-LOOP NORMALIZED STATE LOSS AND ACTION EFFORT. VALUES ARE MEAN ± SD ACROSS EIGHT SEED-LEVEL AGGREGATES. EVERY SEED AGGREGATE GIVES EQUAL WEIGHT TO FIVE SYSTEMS, THREE CONDITIONS, AND FOUR EPISODES PER CELL.

Controller	State loss	Action effort	ID clean	ID corrupt	OOD corrupt
Nominal, last state	0.372 ± 0.089	0.448 ± 0.013	0.362 ± 0.099	0.368 ± 0.098	0.385 ± 0.093
Filter + nominal	0.370 ± 0.090	0.398 ± 0.013	0.362 ± 0.099	0.364 ± 0.099	0.383 ± 0.092
PGNN-style	0.375 ± 0.091	0.464 ± 0.017	0.367 ± 0.100	0.370 ± 0.099	0.388 ± 0.094
CFR-WM	0.370 ± 0.090	0.404 ± 0.016	0.363 ± 0.099	0.364 ± 0.099	0.384 ± 0.092

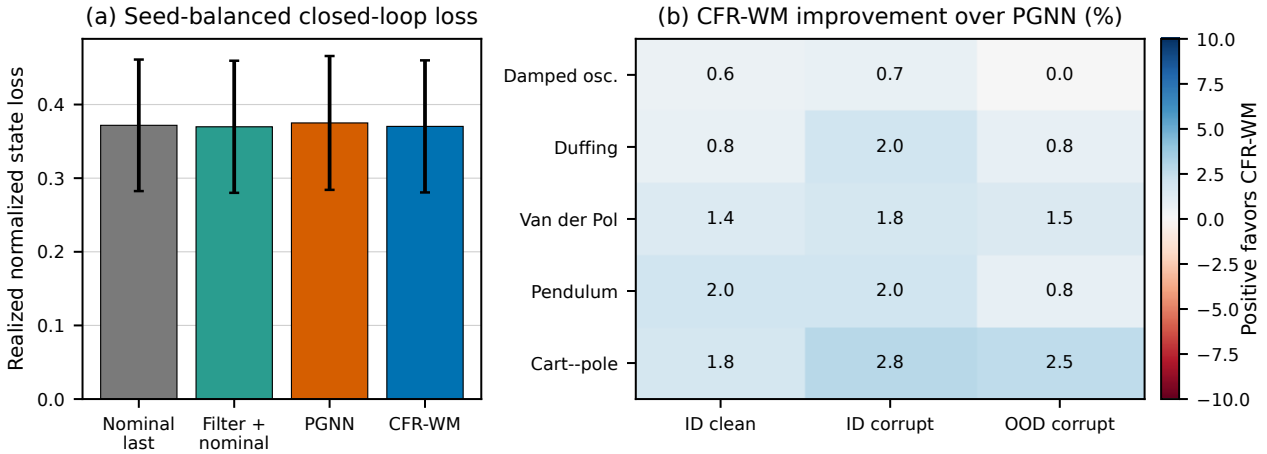


Fig. 1. Closed-loop aggregate and heterogeneity. Error bars in (a) are SD across eight seed-level aggregates; heat-map entries in (b) are seed means.

TABLE VIII
EXTERNAL FULL-STATE NRMSE, MEAN ± SD ACROSS TRAINING SEEDS AFTER EQUAL WEIGHTING OF THE THREE HELD-OUT FLIGHTS.

Context	Horizon	CFR-WM	Unfiltered	Recurrent	Nominal
As recorded	25	0.389 ± 0.012	0.376 ± 0.010	0.388 ± 0.008	0.488
As recorded	50	0.530 ± 0.016	0.518 ± 0.019	0.519 ± 0.006	0.790
Noise	25	0.393 ± 0.012	0.387 ± 0.010	0.398 ± 0.008	0.498
Noise	50	0.532 ± 0.017	0.523 ± 0.018	0.527 ± 0.006	0.797
Missing	25	0.397 ± 0.011	0.386 ± 0.008	0.395 ± 0.005	0.495
Missing	50	0.537 ± 0.014	0.527 ± 0.020	0.526 ± 0.007	0.792
Noise + missing	25	0.401 ± 0.011	0.397 ± 0.008	0.405 ± 0.005	0.506
Noise + missing	50	0.540 ± 0.014	0.534 ± 0.019	0.533 ± 0.006	0.799

TABLE IX
EXTERNAL POSITION AND VELOCITY RMSE IN PHYSICAL UNITS.

	Position RMSE (m)		Velocity RMSE (m/s)	
Context	CFR-WM	Unfiltered	CFR-WM	Unfiltered
As recorded	0.078 ± 0.003	0.083 ± 0.010	0.249 ± 0.015	0.233 ± 0.009
Noise	0.091 ± 0.003	0.115 ± 0.007	0.257 ± 0.014	0.258 ± 0.007
Missing	0.081 ± 0.005	0.095 ± 0.022	0.259 ± 0.014	0.243 ± 0.009
Noise + missing	0.094 ± 0.004	0.125 ± 0.017	0.267 ± 0.014	0.268 ± 0.009

TABLE X
MEASURED-FLIGHT UNCERTAINTY DIAGNOSTICS, AVERAGED EQUALLY OVER HELD- OUT FLIGHTS AND THEN OVER TRAINING SEEDS.

Condition	Method	NLL	Cov.	Width	Cal. error
Recorded	CFR-WM	2.264	0.608	0.453	0.178
Recorded	Unfiltered residual	0.794	0.716	0.526	0.099
Noise	CFR-WM	2.994	0.524	0.454	0.242
Noise	Unfiltered residual	2.361	0.565	0.528	0.217
Missing	CFR-WM	1.829	0.667	0.534	0.123
Missing	Unfiltered residual	0.735	0.779	0.685	0.053
Combined	CFR-WM	2.200	0.613	0.535	0.184
Combined	Unfiltered residual	1.130	0.703	0.688	0.124

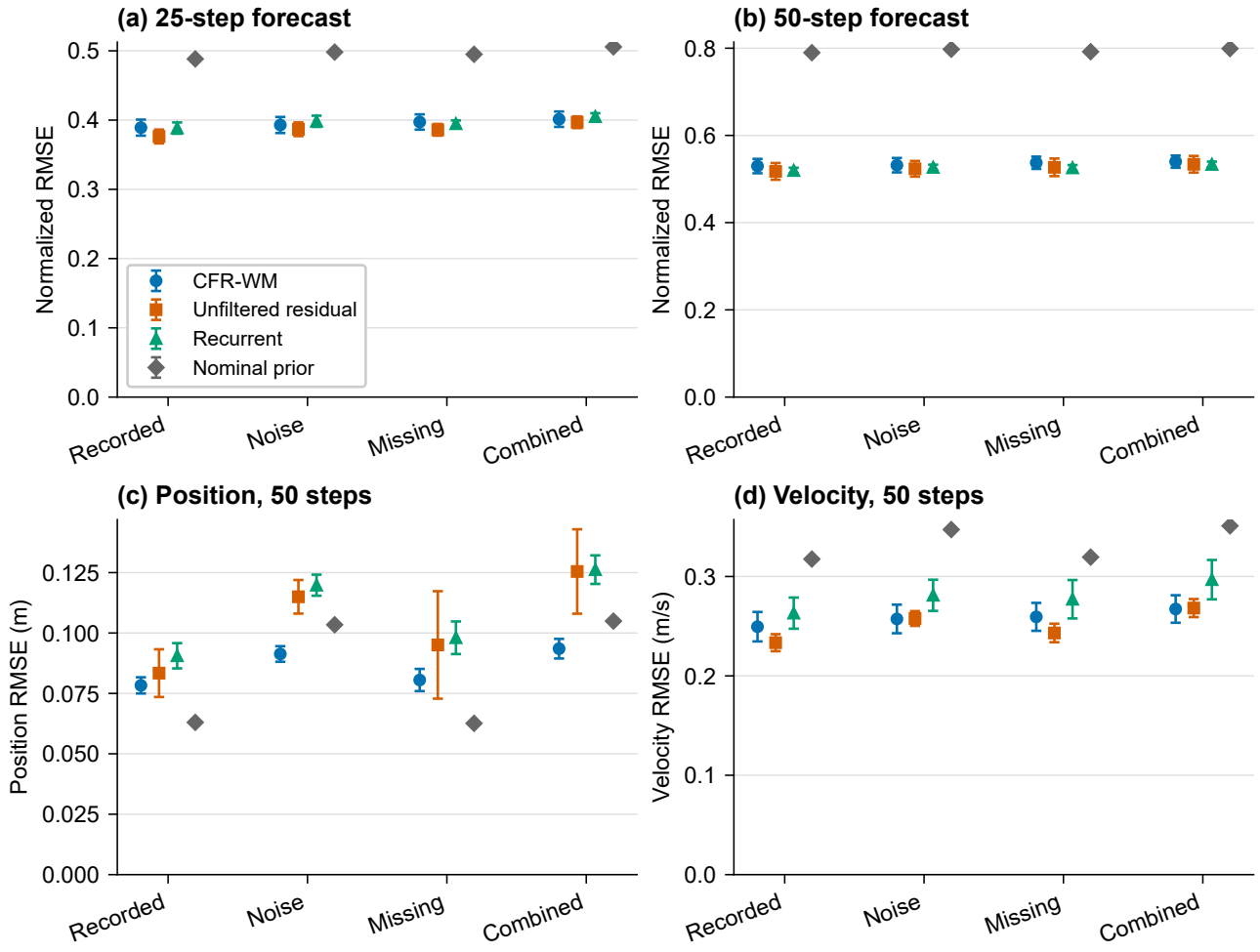


Fig. 2. Measured-flight forecast errors by method, context condition, and horizon. Position and full-state errors need not rank models identically.